

JANMESH JOSHI

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[Github.com/JanmeshJ](#) | [Portfolio](#) | [LinkedIn/janmeshjoshi](#)

PROFESSIONAL SUMMARY

ML & AI Engineer with 4+ years shipping production systems across RAG pipelines, semantic search, and MLOps. Founder of MyNextRoom, a cross-platform rental marketplace with ML-driven fraud detection and 200+ pre-launch signups. MSc in Human-Centered AI (TU Dublin) specialising in Explainable AI and Governance for high-stakes, regulated environments.

TECHNICAL SKILLS

ML & Deep Learning: TensorFlow, PyTorch, Scikit-learn, RAG, Hugging Face, Langchain

LLMs/NLP: GPT-4o, Claude, BERT, BERTopic, UMAP, BGE-M3, Sentence-Transformers, FAISS **MLOps, Cloud &**

Monitoring: AWS, Docker, Google Cloud Run, CI/CD, Git

API & Backend: FastAPI, REST APIs, Python, SQL

EXPERIENCE

Founder & Product Engineer

January 2026 – Present

MyNextRoom | [Link](#)

Dublin, Ireland

- Sole founder and engineer of a cross-platform iOS/Android rental marketplace built with React Native/Expo, Supabase (PostgreSQL), real-time messaging, push notifications, and universal deep linking; submitted to the App Store with 200+ waitlist signups prior to launch.
- Trained and deployed a Random Forest fraud detection classifier on listing price-to-area feature vectors, achieving automated anomaly scoring for real-time listing moderation served via FastAPI on Google Cloud Run.
- Built a k-NN compatibility matching system using Scikit-learn on structured renter-listing feature vectors spanning budget, move-in date, and housing preferences, replacing rule-based filtering with a learned similarity model.
- Integrated GPS-based location features via expo-location, video verification for listing authenticity, and PostHog for product analytics and funnel tracking.

Machine Learning Engineer

June 2020 – Aug. 2024

Panghat

Mumbai, India

- Improved top-5 semantic retrieval relevance by 35% over BM25 baseline across 500K+ legal documents using DPR-CTX and BGE embeddings.
- Built and deployed containerized ML inference services on Docker and Kubernetes, serving 1000+ daily requests with sub-100ms latency.
- Reduced vector search latency by 10x by HNSW index tuning and optimized partitioning strategies in Milvus.
- Developed a RAG-based legal search assistant using LangChain and GPT-4, achieving 92% answer relevance in internal evaluations.

Teaching Assistant (TA)

Feb. 2025 – Present

Technological University of Dublin

Dublin, Ireland

- Delivered Machine Learning lectures (supervised learning, model evaluation, and neural networks, etc.) to 50+ first-year undergraduate students at TU Dublin.
- Taught Python programming including data structures, OOP, and clean code principles, conducting code reviews to enforce modular design and best practices.

PROJECTS

Ethical AI & Risk Modeling for AI Tutors (MSc Thesis)

Sept 2024 - Oct 2025

Technological University of Dublin

Dublin, Ireland

- Engineered a Machine Learning pipeline to detect user over-reliance on AI systems, achieving a Macro F1 score of 0.89 using Random Forest Classifiers.
- Implemented Explainable AI (XAI) techniques using SHAP (Shapley Additive exPlanations) to interpret model decisions, identifying key behavioral drivers like prompt verbosity and solution-seeking patterns.

EDUCATION

Technological University of Dublin

Sept. 2024 – Oct. 2025

Master's, Human Centered Artificial Intelligence

Dublin, Ireland

University of Mumbai

Aug. 2020 – May 2022

Master's, Information Technology

Mumbai, India